MIL-T-60171(MU)
10 March 1965
SUPERSEDING
FA(FC)-PD-86
2 July 1961

MILITARY SPECIFICATION

TAPE READER: 10526192

1. SCOPE

1.1 This specification covers one type of tape reader capable of handling two levels of tape inputs for entering data into the Gun Direction Computer: M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1220

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DRAWINGS

U.S. Army Munitions Command

F10526192 Tape Reader

PACKAGING DATA SHEET

MIL-P-14232/P10526192 Packaging of Tape Reader

(Copies of specifications, standards, drawings and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

Electronic Industries Association (EIA)
Title: Tape, 1 inch perforated
Standard: RS-227
(Applications for copies should be addressed to Electronic Industries Association Engineering Office, 11 W. 42nd St. New York, 36 N.Y.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Connectors.- Connectors functioning as input/output (interface) terminations shall be as designated in Drawing F10526192.

3.1.2 Hardware.- Stainless steel fasteners (bolts, screws. nuts, washers, etc.) shall be utilized throughout the construction of the equipment.

3.1.3 Solid state devices.- The equipment design shall make full use of solid state devices, where their use will aid in accomplishing the requirements of weight, size, reliability, power, cooling, etc. However, the use of solid state devices shall not alter the equipment in such a manner that it will not meet the requirements of this specification.

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3.2 Design and construction.- The tape reader shall be designed and constructed in accordance with Drawing F10526192 of this specification. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.2.1 Fire hazard.- The design and construction of the equipment shall be such that its use shall not, under any conditions of storage or operation, create a fire hazard.

3.2.2 Special tools.- The equipment shall require no special tools for assembly, disassembly or maintenance.

3.2.3 Mounting attitude.- The equipment shall operate satisfactorily when mounted in any position.

3.2.4 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in manufacturer's part numbers.

3.2.5 Reference designations.- Reference designations shall be in accordance with MIL-STD-16. The unit numbering method described in MIL-STD-16 shall apply. The assignment of reference designations ahll be made on schematic diagrams drawn in conformance with MIL-D-70327.

3.2.6 Finishing.- The finishing procedure shall conform to the general specification specified in 3.3.1 of this specification with the exception that all external metallic surfaces shall be hard anodized, unless otherwise specified.

3.2.7 Size and mounting.- The equipment shall conform to the configuration shown in Drawing F10526192.

3.2.8 Weight.- The weight of the equipment shall not exceed two and one-half pounds.

3.2.9 Tape.- The tape reader shall be capable of handling two types of tape inputs:

(a) Eight level tape
(b) Five level tape

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3.2.9.1 Eight level tape.- Eight level tape is defined as tape conforming to EIA Standards proposal of 26 September 1958, entitled "Final Proposal of Standard One Inch Perforated Paper Tape." This proposal is presented as follows:

"Final proposal of standard one inch perforated paper tape".

1.0 Intended to serve as a guide in the coordination of equipment design.
1.1 This standard is for perforated paper tape with fully punched round holes.
2.0 Standard one inch perforated paper tape.
2.1 The unpunched tape should have an overall width of 1.00 inch plus or minus 0.003 inch after the tape has been conditioned to 73 plus or minus 3.5 degrees Fahrenheit and 50 percent, plus or minus 2 percent, relative humidity for a period of 24 hours.
2.21 The tape should be used for recording six, seven, or eight levels of information across the tape.
2.3 When the tape is edge guided, the guided edge of the tape should be parallel with the longitudinal centerline of the feed holes centerline on the three-level side of the tape.
2.4 The feed holes in the tape should be round with a diameter of 0.045 inches, plus 0.002, minus 0.001 in.
2.5 The code holes in the tape should be round, with a diameter of 0.072 inches, plus 0.001, minus 0.002 inches.
2.6 All holes punched in the tape should nominally center on the true intersection of longitudinal and perpendicular transverse center lines spaced 0.100 inches apart. Tolerance on locations of code lines in any one transverse row, relative to the centerlines of the feed hole in that row, should be plus or minus 0.002 inches in the transverse direction and plus or minus 0.003 inches in the longitudinal direction. Tolerance on distance between centers of adjacent feed holes should be plus or minus 0.001 inches. Accumulated longitudinal errors between centers of feed holes should not exceed plus or minus 0.009 inches within spans of 0.9 inches to 6 inches.
2.7 The measurement of all dimensions of the punched paper tape should be made with the punched paper tape conditioned to the same environmental conditions under which it was punched.

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3.2.9.2 Feed holes, location of.- The feed holes shall be located with their centers 0.392 $^{+0.005}_{-0.003}$ inches from the reference or guided edge of the tape.

3.2.9.3 Five level tape.- Five level tapes shall be of two types:

(a) Type I - 11/16 inches wide
(b) Type II- 7/8 inches wide.

3.2.9.3.1 Type I tape.- Type I tape shall identical to and shall conform to the requirements of eight level tape except as follows:

(a) Width - The width of Type I tape shall be 0.687 ± 0.003 inches.
(b) Hole capacity - The hole capacity of each transverse row shall be five code holes.
(c) Feed hole, location of - See paragraph 3.2.9.2.

3.2.9.3.2 Type II tape.- Type II tape shall be identical to and shall conform to the requirements of eight level tape except as follows:

(a) Width - The width of Type II Tape shall be 0.875 ± 0.003 inches.
(b) Hole capacity - The hole capacity of each transverse row shall be five code holes.
(c) Feed hole, location of - The feed holes shall be located with their centers 0.578 ± 0.003 inches from the reference or guided edge of the tape.

3.2.9.4 Reference or guided edge.- In all cases, the reference or guided edge is defined as that edge of the tape toward which the grouping of three code holes may appear on the tape.

3.2.9.5 Tape thickness.- The equipment shall be capable of accepting tapes ranging in thickness from 0.002 inches to 0.010 inches including splices.

3.2.9.6 Folded or creased tape.- The equipment shall be capable of reading not only rolled tapes, but also tapes that have been fan-folded, (accordion pleats every 8-1/2 inches).

3.2.9.7 Tape take up and supply.- No provision shall be included on this tape reader for tape takeup or supply reels.

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3.2.10 Tape guides.- Provisions shall be made for guiding the tape sufficiently to prevent the tape from skewing while being read.

3.2.10.1 Adjustable tape guides.- The tape guides shall be made adjustable to accommodate all three sizes of tapes mentioned in 3.2.9.1 and 3.2.9.3; namely, the eight level and the two types of five level tapes. The adjustment shall be able to be made without the use of tools.

3.2.11 Interlocking device.- An interlocking device shall be provided to prevent the equipment from driving or reading tape when the equipment runs out of tape or while it is being loaded.

3.2.12 Drag mechanism.- The sprocket detent shall slip, to prevent driving of the tape, whenever the drag of the tape is eight (8) ounces minimum to twelve (12) ounces maximum."

3.2.13 Tape life.- Code sensing and tape driving mechanisms shall be such as to assure that common 0.003 inch thick paper tapes shall have a ninety percent life expectancy in excess of five hundred passes.

3.2.14 Arc supression.- The arc suppression circuit, shunted across the solenoid, shall limit inductive voltage spikes to 85 volts peak.

3.2.15 Interface connector terminations.- The interface connectors as shown in Drawing F10526192 shall be wired in accordance with Figure 2.

3.3 General specification.- The tape reader shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.3.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship
e. Soldering electrical connections

3.3.2 MIL-C-46357

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes
e. Handling

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3.4 Environmental service conditions.

3.4.1 Vibration.- The tape reader shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the tape reader for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the tape reader shall not be damaged or adversely affected and shall meet the requirements of 3.5.

3.4.2 Shock.- The tape reader shall withstand 3 shock impulses in each of 3 mutually perpendicular directions. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.4.3 Temperatures.

3.4.3.1 Operating.- The tape reader shall meet the requirements of 3.5.1, 3.5.3(b), & 3.5.4 while exposed to temperatures of -40°F and +135°F.

3.4.3.2 Storage.- The tape reader shall meet the requirements of 3.5 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.4.4 Humidity.

3.4.4.1 Operating.- The tape reader shall meet the requirements of 3.5.1, 3.5.3(b), & 3.5.4 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.4.4.2 Non-operating.- The tape reader shall meet the requirements of 3.5 after exposure to (and when thoroughly dired) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.4.5 Fungus resistance.- The tape reader shall show no evidence of damage and shall meet the requirements of 3.5, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

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3.4.6 Salt spray.- The tape reader shall show no evidence of physical damage and meet the requirements of 3.5 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93°and 97°F.

3.4.7 Sand and dust.- The tape reader, simulated in its computer mounting positon and dust cover latched in place, shall withstand 6 hours of exposure to an atmosphere heavily laden with sand and dust particles at an ambient temperature of 160°F. Subsequent to the exposure and with dust cover removed, there shall be no evidence of sand or dust and shall meet the requirements of 3.5.

3.5 Performing.- Unless otherwise specified the tape reader shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%.

3.5.1 Input power.- The equipment shall operate from the following power source:

> 31 volts dc ± 7 volts, maximum ripple of 30% peak to peak. Current shall not exceed 2.42 amperes peak at any operating temperature.

3.5.2 Power consumption.- The maximum average power consumed by the reader at its free running reading speed shall be twenty watts (sixty-eight watts peak) at the nominal supply voltage and at 72°F.

3.5.3 Operating speed.

(a) The free running speed at 72°F and nominal voltage input shall be within the range of from 20 to 35 code lines per second.

(b) The free running speed with any combination of environmental or voltage requirements shall be within the range of from 15 to 45 code lines per second.

3.5.4 Output pulse.- The tape reader shall display output pulses as shown in Figure 1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Examinations and tests related to Section 3 herein shall be performed on a deflect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Tables I to IV inclusive. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be conducted at the standard ambient temperature (60° to 90°F).

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TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level I of Table I with
Sampling Plan Table II-A of MIL-STD-105

		Requirement	Test Procedure
CRITICAL:	None		
MAJOR:	AQL 1.0 percent defective		
101.	Input power	3.5.1	4.5.3.1
102.	Power consumption	3.5.2	4.5.3.3
103.	Operating speed	3.5.3	4.5.3.2
104.	Output pulse	3.5.4	4.5.3.4
MINOR:	None		

4.3 Standard sampling.

4.3.1 General.- One unit shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table II after having successfully met those requirements and tests in Table I.

TABLE II

Characteristic		Requirement	Test Procedures
301.	Connectors	3.1.1	F10526193, Visual
302.	Hardware	3.1.2	Visual
303.	Reference designations	3.2.5	MIL-STD-16, Visual
304.	Finishing	3.2.6	MIL-F-13926
305.	Size and mounting	3.2.7	4.5.2.7
306.	Weight	3.2.8	4.5.2.8
307.	Tape guides	3.2.10	4.5.2.2
308.	Tape life	3.2.13	4.5.2.6
309.	Interface connector terminations	3.2.15	Visual
310.	General specifications	3.3	MIL-F-13926 and MIL-C-46357, visual

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4.3.1.1 Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The standard samples shall have met the requirements of Table I and shall then meet the requirements and tests in Table III.

TABLE III

Characteristic	Requirement	Test Procedure
311. Mounting attitude	3.2.3	4.5.2.5
312. Tape	3.2.9	4.5.2.1
313. Interlocking	3.2.11	4.5.3.1(a)
314. Drag mechanism	3.2.12	4.5.2.3
315. Arc suppression	3.2.14	4.5.2.4

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table IV.

TABLE IV

Characteristic	Requirement	Test Procedure
303. Vibration	3.4.1	4.5.1.1
304. Shock	3.4.2	4.5.1.2
305. Humidity-operating	3.4.4.1, 3.5.1 3.5.3(b) 3.5.4	4.5.1.2.3, 4.5.3.1 4.5.3.2, 4.5.2.4
306. Humidity-non-operating	3.4.4.2	4.5.1.3.1
307. Salt spray	3.4.6	4.5.1.4
308. Operating temperature	3.4.3.1, 3.5.1 3.5.3(b), 3.5.4	4.5.1.5, 4.5.3.1, 4.5.3.2, 4.5.3.4
309. Storage temperature	3.4.3.2	4.5.1.5
310. Sand and dust	3.4.7	4.5.1.7

4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the representative inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

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4.3.4 Certification.- The requirements in Table V below shall be considered for acceptance by certification. Certification shall include test data and results for all characteristics specified above. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor.

TABLE V

Characteristic	Requirement
316. Solid state devices	3.1.3
317. Fire hazard	3.2.1
318. Special tools	3.2.2
319. Interchangeability	3.2.4

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

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Item Number	Description
1 - Voltage measuring device	0-50 Vdc, input resistance 10 megohms or greater, accuracy of ± .5% or better
2 - Current measuring device	0 to 5 amperes, accuracy of ± .5% full scale
3 - Oscilloscope	Sweep time of scope 1 millisecond per centimeter, input resistance 8 megohms
4 - Power supply	Plus 0-38 Vdc, maximum ripple 30% peak to peak, current rating 2.5 amperes

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.4.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.4.1 with the unit rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.5.

4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.4.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5G's. The unit shall be subsequently inspected for physical damage or defects in compliance with 3.4.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the requirements of 3.5 as required by 3.4.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.4.4.2 and the humidity test

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defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall than be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.5 as required by 3.4.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.4.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.3.1, 4.5.3.2 and 4.5.3.4 to determine compliance with 3.4.4.1.

4.5.1.4 Salt spray.- The unit shall be subjected to the conditions in 3.4.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.3 to 4.5.3.4 inclusive for compliance with 3.4.6.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the unit shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.4.3 shall apply. While at the operating temperatures of 3.4.3.1, the unit shall be tested in accordance with 4.5.3.1, 4.5.3.2 and 4.5.3.4 to determine compliance with the requirements of 3.5.1, 3.5.3(b) and 3.5.4 as required by 3.4.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.3 to 4.5.3.2, inclusive to determine compliance with 3.4.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.4.5, one of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.1.7 Sand and dust.- This test shall be conducted in accordance with the sand and dust test defined in MIL-F-13926 to determine compliance with 3.4.7. Temperature changes shall not occur at a rate exceeding 2 degrees Fahrenheit per minute to avoid

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thermal shock. The unit shall be positioned in the same manner as it is positioned when functioning as a part of Computer Set, Gun Direction: M18. Operation while at 77°F shall be conducted in accordance with the tests in 4.5.3 to 4.5.3.4 inclusive.

4.5.2 Design and construction.

4.5.2.1 Tape.- The tape reader shall be subjected to one complete cycle of 1000 foot length of fan folded tape for each of three widths of tape specified in 3.2.9.1 and 3.2.9.3. The operation of the tape reader shall be visually observed throughout the entire test to determine compliance with the requirements of 3.2.9.1 to 3.2.9.7.

4.5.2.2 Tape guides.- This test shall be conducted concurrent with the test specified in 4.5.2.1. The tape guides shall be subjected to a visual and tactile inspection during the operation of the tape reader to determine compliance with the requirements of 3.2.10 and 3.2.10.1.

4.5.2.3 Drag mechanism.- The drag mechanism shall be tested for slippage under normal operation by means of a weight attached to the tape. Slippage shall occur within the limits specified in 3.2.12.

4.5.2.4 Arc suppression.- This test shall be conducted with standard measuring equipment. Voltage limit shall conform to the requirement specified in 3.2.14.

4.5.2.5 Mounting attitude.- To determine compliance with 3.2.3 the tape reader shall be operated as specified in 4.5.3.1 in each of three mutually perpendicular positions. The tape reader shall operate successfully for a minimum of 300 feet in each position without failure.

4.5.2.6 Tape life.- Prior to final acceptance one tape reader shall be randomly selected from the first lot of each contract or purchase order and shall be operated for a period of 500 hours, during which time it shall read a minimum of 500,000 feet of tape without major overhaul, and only with normal maintenance. Upon completion of the life test, the functional test described in paragraph 4.5.3 shall be conducted to determine compliance with 3.2.13.

4.5.2.7 Size and mounting.- The tape reader shall be subjected to a visual and dimensional inspection to determine compliance with the requirements specified in 3.2.7. Dimensional measurements shall be made by means of standard measuring equipment.

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4.5.2.8 Weight.- The weight of the tape reader shall be measured with standard measuring equipment. The weight shall not exceed the limit specified in 3.2.8.

4.5.3 Performance.- The functional schematic shown in Figure 2 is for reference only. The designated contacts and symbols in the following tests are shown in this schematic only as a guide in meeting the specified requirements.

4.5.3.1 Input power.- To determine compliance with 3.5.1 the input power shall drive the actuating solenoid in order to operate the tape reader under the following conditions with contacts "RTC" and "A" in the normally closed position.

(a) Contact RTC is transferred to the normally open position when loading tape, or no tape, or end of tape.

(b) Contact A is transferred to the normally open position when there is an excessive amount of tension in the tape.

4.5.3.1.1 Outputs.- The following outputs shall be provided in the form of single pole, double throw contacts.

(a) Contacts RC1 through RC8 - Information contacts (8)- The minimum period of transfer when reading a hole shall be two milliseconds.

(b) Contact RCC - Feedhole or Commont contact.- The minimum period of transfer when reading a feedhole shall be one millisecond.

(c) Contact B - End of cycle contact - Operation of this contact is described in 4.5.3.1.2(d).

4.5.3.1.2 Contact timing.- For continuous cycling of the equipment, the following conditions on contact sequencing and drive signal application shall apply.

(a) Starting (Refer to figure 1) - The start of a reading cycle shall be defined as the time of application of the input drive signal (see 4.5.3.1).

(b) Contact RCC - The closing of contact RCC will be used externally to the equipment to de-energize the input drive signal. Inertial action of the equipment shall guarantee a minimum period of transfer as specified in 4.5.3.1(b).

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(c) Contacts RC1 through RC8 - Contacts RC1 through RC8 when reading a hole, shall transfer prior to the transfer of contact RCC and shall return subsequent to the return of contact RCC.

(d) Contact B - The transfer of contact B during the input drive portion of one read cycle shall occur prior to the transfer of any of the information contacts. During the return stroke, the return of contact B will be used to an output to enable the input drive signal if another character is to be read. Therefore, a complete mechanical cycling of reading one character including one character tape advance shall be completed prior to the return of Contact B.

4.5.3.1.3 Stopping.- Stopping shall be provided externally to the equipment by the absence of an input drive signal following the return of contact B (see 4.5.2.1.2(d))

4.5.3.1.4 Contact rating

4.5.3.1.4.1 Contacts RTC and A.- RTC contacts shall be rated at 5 amperes at 250 volts AC, and "A" contact rated at 2 amperes. Contact resistance shall be less than 0.1 ohm at 1 ampere."

4.5.3.1.4.2 Other contacts.- All other contacts shall be capable of carrying 100 milliamperes into a resistive load. Contact resistance shall be 0.15 ohms or less when measured from the connector.

4.5.3.2 Operating speed.- With the tape reader operating as specified in 4.5.3.1, determine compliance with 3.5.3 by measuring the free running speed using a premeasured length of tape. Measure the length of time the tape passes through the tape reader by means of standard measuring equipment. The free running speed shall be within the limits of 3.5.3(a) and (b).

4.5.3.3 Power consumption.- Measure the outputs of the tape reader while operating the tape reader as specified in 4.5.3.2. The output of the tape reader shall not exceed the limits of 3.5.2.

4.5.3.4 Output pulse.- Operate the tape reader as specified in 4.5.3.1. The tape reader shall exhibit the output pulses shown in Figure I in compliance with 3.5.4 when subjected to the following conditions:

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(a) Contact RCC.- Contact RCC shall be closed for a minimum time of one millisecond.

(b) Information contacts (RC1 through RC8) - The information contacts RC1 through RC8 shall be closed when reading a hole for a minimum time of two milliseconds. This transfer period of all information contacts shall be longer than the transfer period of contact RCC as shown in Figure I.

(c) Contact bounce (see Figure I) - There shall be no bounce on contacts B or B' during the period when contact RCC is transferred.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10526192.

6. NOTES

6.1 Intended use.- The tape reader shall provide the capacity of reading input data as punched on three different widths of tape and entering the data into a computer.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of packaging and packing required
(c) Applicable stock number
(d) Applicable Packaging Data Sheet (see 5.1)
(e) A block of serial numbers to identify each tape reader

Custodian:

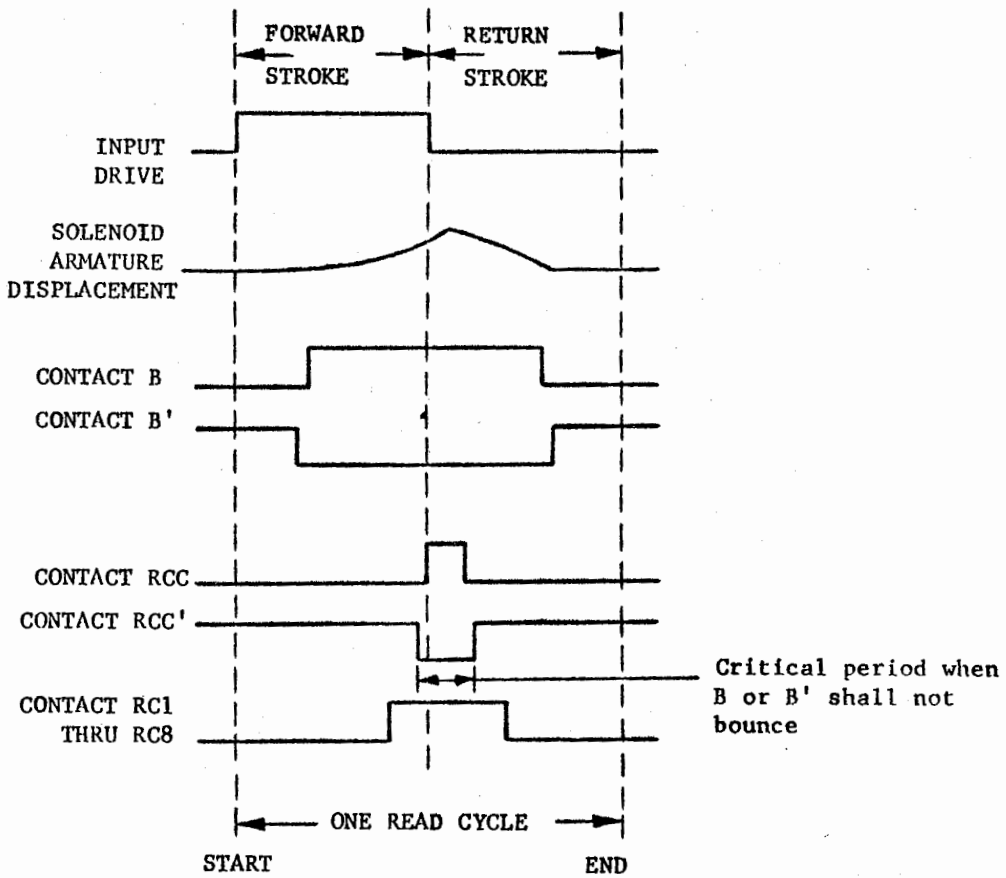
Army - MU

Preparing activity:

Army - MU

Project No. 1220-A208Y

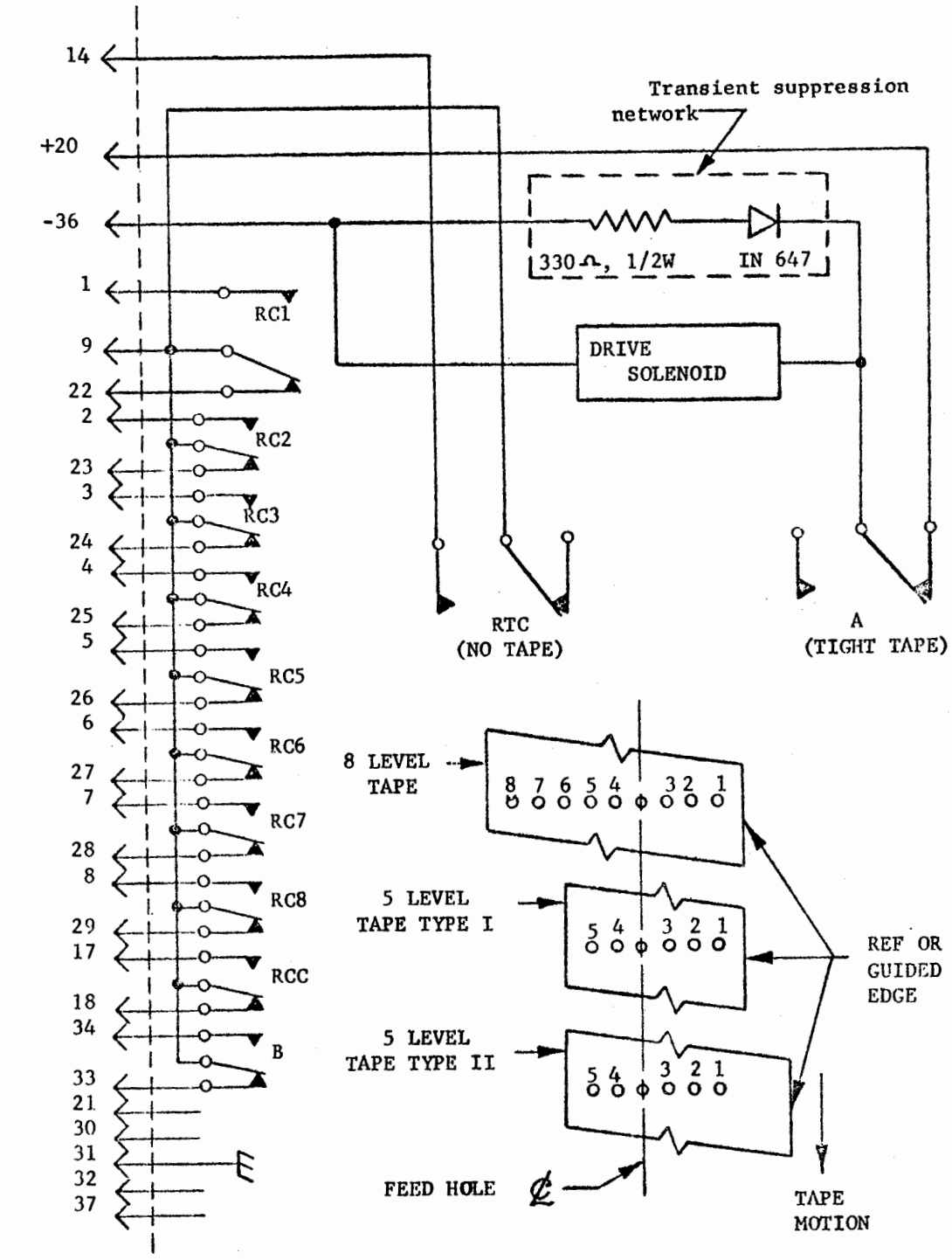
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CONTACT SEQUENCE CHART

FIGURE 1

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P1 DCM 37-P
Cannon (or equiv.)

CIRCUIT DIAGRAM

FIGURE 2

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